```
clc;
clear;
close all;

%% ============================================================
% IIR FILTER REALIZATION AND COMPARATIVE ANALYSIS
% Direct Form-I, Direct Form-II, Cascade and Parallel
%% ============================================================

%% 1. IIR FILTER COEFFICIENTS

b = [0.0048 0.0193 0.0289 0.0193 0.0048];
a = [1 -2.3695 2.3140 -1.0547 0.1874];

N = 50;                         % Number of samples
x = [1 zeros(1, N-1)];          % Unit impulse input
n = 0:N-1;                      % Sample index

%% 2. DIRECT FORM-I AND DIRECT FORM-II REALIZATIONS

% Direct Form-I
y1 = filter(b, a, x);

% Direct Form-II (canonical form)
y2 = filter(b, a, x);

%% 3. CASCADE REALIZATION

% Convert transfer function to second-order sections
[sos, g] = tf2sos(b, a);

% Apply overall gain
yc = g * x;

% Filter through each second-order section
for k = 1:size(sos, 1)
    yc = filter(sos(k, 1:3), sos(k, 4:6), yc);
end

%% 4. PARALLEL REALIZATION

% Partial fraction expansion
[r, p, k] = residuez(b, a);

% Initialize output using the direct term
yp = filter(k, 1, x);

% Add the first-order sections
for i = 1:length(r)
```

```
    yp = yp + filter(r(i), [1 -p(i)], x);
end

%% 5. DISPLAY FILTER COEFFICIENTS

fprintf('\n=====================================================\n');
```

```
=====================================================
```

```
fprintf('             IIR FILTER COEFFICIENTS\n');
```

```
            IIR FILTER COEFFICIENTS
```

```
fprintf('=====================================================\n');
```

```
=====================================================
```

```
disp('Numerator coefficients b =');
```

```
Numerator coefficients b =
```

```
disp(b);
```

```
    0.0048    0.0193    0.0289    0.0193    0.0048
```

```
disp('Denominator coefficients a =');
```

```
Denominator coefficients a =
```

```
disp(a);
```

```
    1.0000   -2.3695    2.3140   -1.0547    0.1874
```

```
disp('Second-Order Sections (Cascade) =');
```

```
Second-Order Sections (Cascade) =
```

```
disp(sos);
```

```
    1.0000    2.1551    1.0000    1.0000   -1.0486    0.2962
    1.0000    1.8657    1.0000    1.0000   -1.3209    0.6327
```

```
disp('Cascade Gain =');
```

```
Cascade Gain =
```

```
disp(g);
```

```
    0.0048
```

```
disp('Residues (Parallel) =');
```

```
Residues (Parallel) =
```

```
disp(r);
```

```
  -0.2698 + 0.1343i
  -0.2698 - 0.1343i
   0.2594 - 0.8024i
   0.2594 + 0.8024i
```

```
disp('Poles (Parallel) =');
```

```
Poles (Parallel) =
```

```
disp(p);
```

```
   0.6604 + 0.4433i
   0.6604 - 0.4433i
   0.5243 + 0.1459i
   0.5243 - 0.1459i
```

```
disp('Direct Term =');
```

```
Direct Term =
```

```
disp(k);
```

```
    0.0256
```

```
%% 6. ERROR AND PERCENTAGE DEVIATION

% Maximum absolute error relative to Direct Form-I
E = [max(abs(y1 - y2)), ...
     max(abs(y1 - yc)), ...
     max(abs(y1 - yp))];

% Percentage deviation relative to the reference response
reference = max(abs(y1));

if reference > 0
    D = E / reference * 100;
else
    D = zeros(size(E));
end

% Include Direct Form-I as the reference
Structures = ["Direct Form-I";
              "Direct Form-II";
              "Cascade";
              "Parallel"];

Maximum_Error = [0; E(:)];
Deviation_percent = [0; D(:)];
```

```
% Create comparison table
T = table(Structures, Maximum_Error, Deviation_percent, ...
    'VariableNames', ...
    {'Structure', 'Maximum_Error', 'Deviation_percent'});

disp(' ');
```

```
disp('==================================================');
```

```
==================================================
```

```
disp('             REALIZATION COMPARISON');
```

```
            REALIZATION COMPARISON
```

```
disp('==================================================');
```

```
==================================================
```

```
disp(T);
```

Structure	Maximum_Error	Deviation_percent
"Direct Form-I"	0	0
"Direct Form-II"	0	0
"Cascade"	1.9221e-15	8.242e-13
"Parallel"	1.9811e-15	8.4949e-13

```
%% 7. IMPULSE RESPONSE COMPARISON

figure;

stem(n, y1, 'filled');
hold on;
stem(n, y2, '--');
stem(n, yc, ':', 'LineWidth', 1.5);
stem(n, yp, '-.');
```

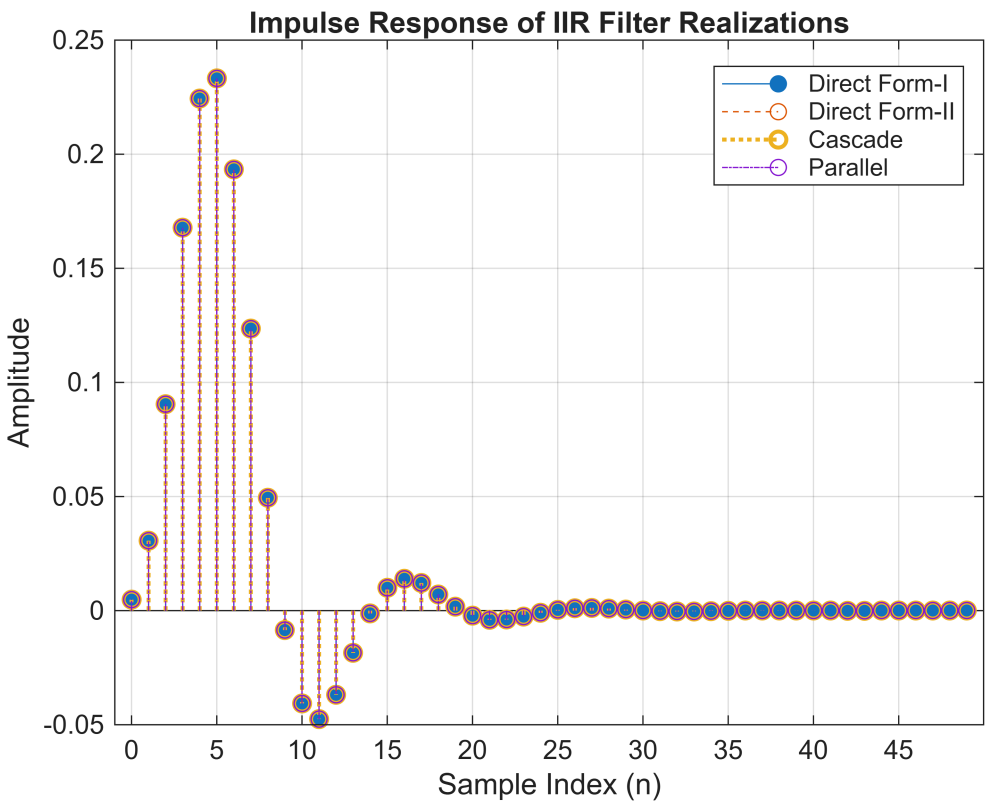
```
Warning: Using only the real component of complex data.
```

```
grid on;
xlabel('Sample Index (n)');
ylabel('Amplitude');

title('Impulse Response of IIR Filter Realizations');

legend('Direct Form-I', 'Direct Form-II', ...
       'Cascade', 'Parallel', ...
       'Location', 'best');
```

```
hold off;
```




```
%% 8. NUMERICAL ERROR COMPARISON

figure;

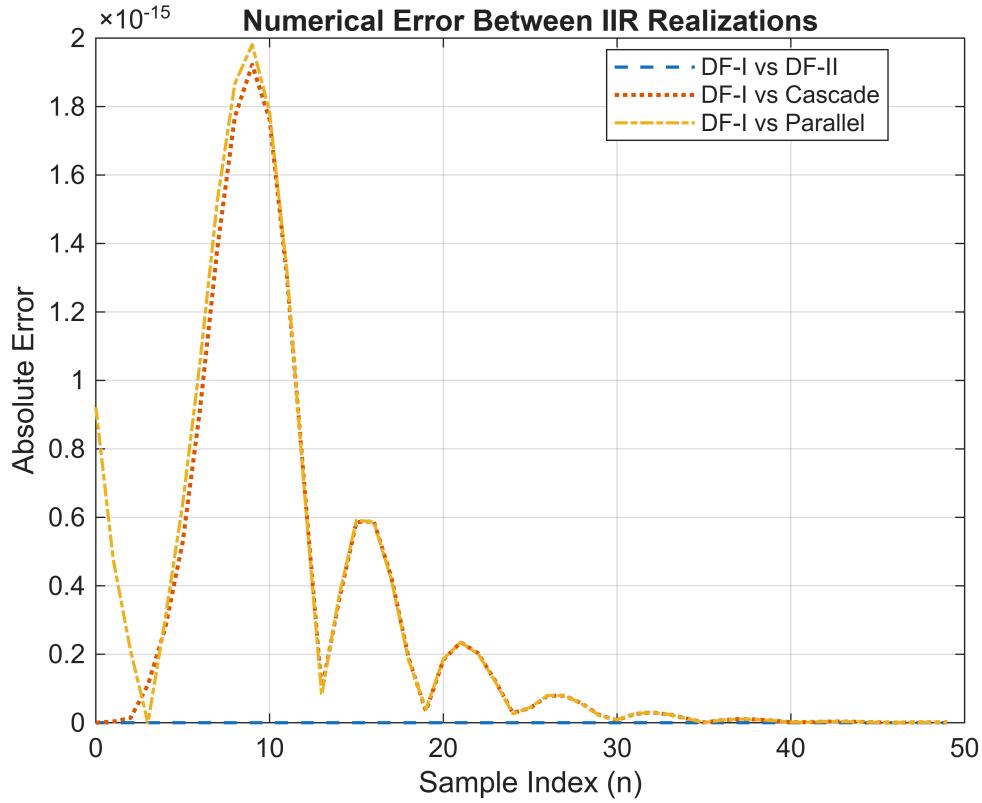
plot(n, abs(y1 - y2), '--', 'LineWidth', 1.2);
hold on;
plot(n, abs(y1 - yc), ':', 'LineWidth', 1.5);
plot(n, abs(y1 - yp), '-.', 'LineWidth', 1.2);

grid on;
xlabel('Sample Index (n)');
ylabel('Absolute Error');

title('Numerical Error Between IIR Realizations');

legend('DF-I vs DF-II', ...
       'DF-I vs Cascade', ...
       'DF-I vs Parallel', ...
       'Location', 'best');

hold off;
```

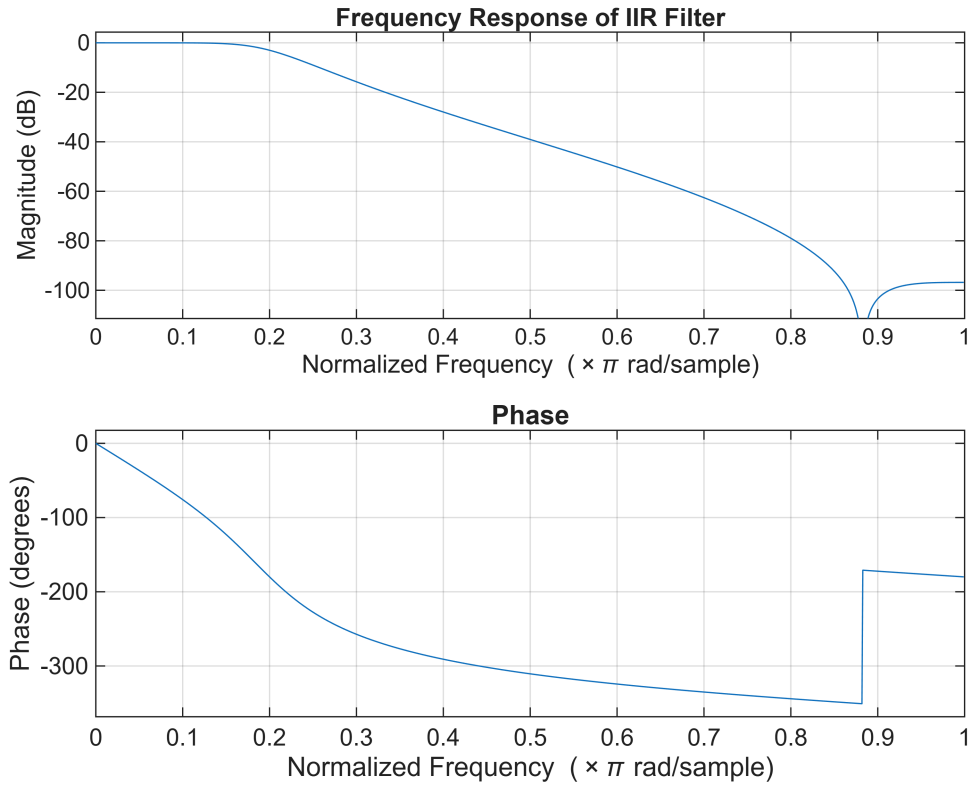



```
%% 9. FREQUENCY RESPONSE

figure;

freqz(b, a, 1024);

title('Frequency Response of IIR Filter');
```

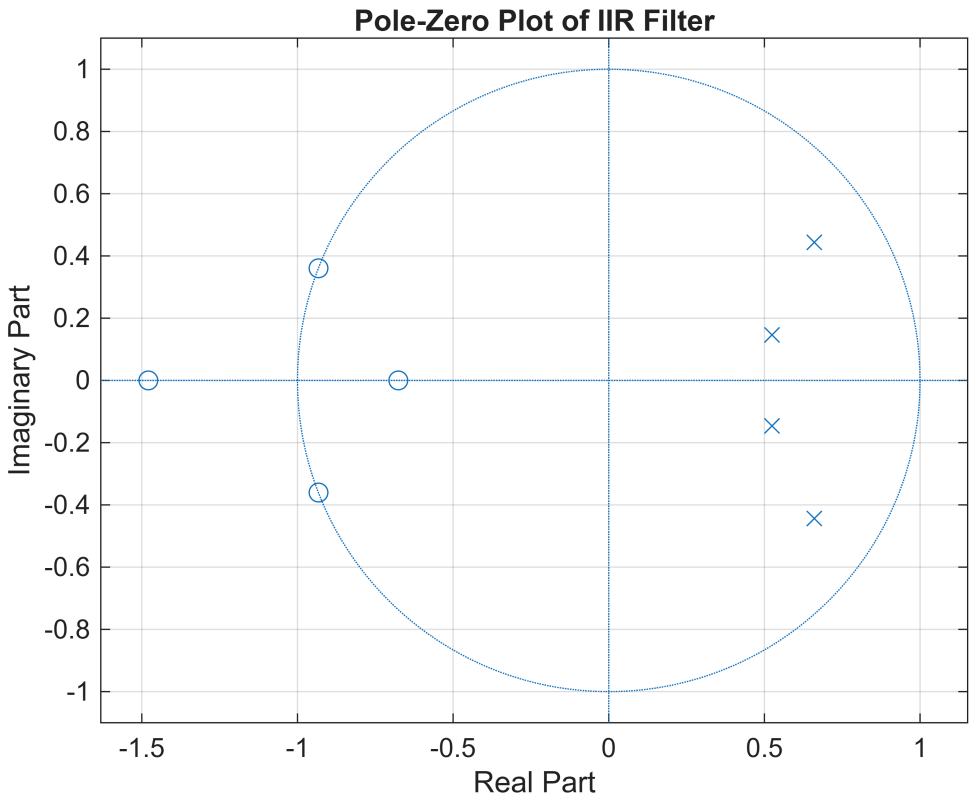



```
%% 10. POLE-ZERO PLOT

figure;

zplane(b, a);
grid on;

title('Pole-Zero Plot of IIR Filter');
```




```
%% ========================================================
% END OF PROGRAM
%% ========================================================
```